Certified Heterogeneous Entropy with Algebraic Randomness Extraction

Daniel Mo Houshmand
QDaria Quantum Research, Oslo, Norway
mo@qdaria.com
ORCID: 0009-0008-2270-5454

***Abstract*—Composing entropy from heterogeneous sources—quantum hardware, WiFi Channel State Information (CSI), and operating system random number generators—is essential for post-quantum cryptographic systems but lacks formal provenance guarantees. We present the Certified Heterogeneous Entropy (CHE) framework, which makes three contributions. First, we introduce Algebraic Randomness Extraction (ARE), a new family of seeded extractors parameterized by algebraic programs over bounded number domains with six arithmetic operations, generated deterministically from SHAKE-256. The core construction operates over five classical domains ($\mathbb{N}_n, \mathbb{Z}_n, \mathbb{Q}_n, \mathbb{R}_n, \mathbb{C}_n$) and extends to quaternions ($\mathbb{H}$, non-commutative), octonions ($\mathbb{O}$, non-associative), finite fields $\mathrm{GF}(p^n)$ (provable per-step uniformity), and p-adic numbers. Second, we describe a certified composition protocol that XOR-fuses entropy from multiple independent sources while constructing Merkle-tree provenance certificates proving per-source contribution. Third, we show that the framework degrades gracefully when individual sources fail, maintaining accurate min-entropy bounds without silent fallback. Our implementation processes 6.8 MB of IBM Quantum entropy (156 qubits), WiFi CSI entropy from ESP32-S3 hardware, and OS entropy through the ARE extractor and compositor pipeline. NIST SP 800-90B testing validates output quality. The provenance certificates provide a complete audit trail satisfying DORA Art. 7 requirements for cryptographic key lifecycle management in EU-regulated financial institutions. We formally prove that GF-domain steps preserve min-entropy exactly (bijection on the multiplicative group) and show empirically that the two-layer architecture (algebraic fold + SHA-256 expansion) provides defense-in-depth extraction distinct from hash-only approaches. We identify what is proven, what is conjectured, and what remains open, providing a clear roadmap for the extractor theory community. All code is open-source and available at https://github.com/QDaria/zipminator.**

1. Introduction

Post-quantum cryptographic systems such as ML-KEM-768 [30] require high-quality entropy for key generation, but production environments rarely depend on a single entropy source. Quantum random number generators (QRNGs) provide information-theoretic guarantees [27] but are expensive and intermittently available. Classical hardware sources such as WiFi CSI [4] offer abundant, zero-marginal-cost entropy but with weaker guarantees. Operating system PRNGs (`os.urandom`) are ubiquitous but opaque. Our prior work on quantum-certified anonymization [3] and unilateral CSI entropy [4] demonstrated the need for multi-source entropy composition. In practice, cryptographic systems must compose entropy from whichever sources are available, yet no existing framework provides cryptographic proof of which sources contributed to a given key's entropy, nor faithfully adjusted degradation bounds when sources fail.

The European Union's Digital Operational Resilience Act (DORA) [16], effective since 1 July 2025, requires financial entities to document cryptographic policies, manage key lifecycles (Art. 7), and respond to cryptographic developments including quantum threats (Art. 6.4). A provenance certificate that answers "which entropy sources contributed to this key, and what was their health status at generation time?" directly addresses these regulatory requirements.

We present the Certified Heterogeneous Entropy (CHE) framework with four contributions:

1) **Algebraic Randomness Extraction (ARE).** A new family of algebraic pre-conditioners operating over nine bounded number domains ($\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}, \mathrm{GF}(p^n), \mathbb{Q}_p$) with six arithmetic operations, composed with SHA-256 for a two-layer extraction architecture. Unlike hash-based extractors (Trevisan [33], GUV [22], leftover hash lemma [23]), ARE provides seed-parameterized algebraic mixing as defense in depth. We prove per-step uniformity for the GF domain (Theorem 8.3) and identify the general extraction bound as an open problem.
2) **Certified composition with Merkle provenance.** An XOR-based composition protocol where each entropy contribution is recorded in a provenance record with canonical encoding. A SHA-256 Merkle tree over these records yields a single root hash that cryptographically binds the composition to its constituent sources.
3) **Graceful degradation with accurate bounds.** When individual sources fail, the compositor excludes them, adjusts the min-entropy estimate downward, and records the degradation in the provenance certificate. No silent fallback occurs.

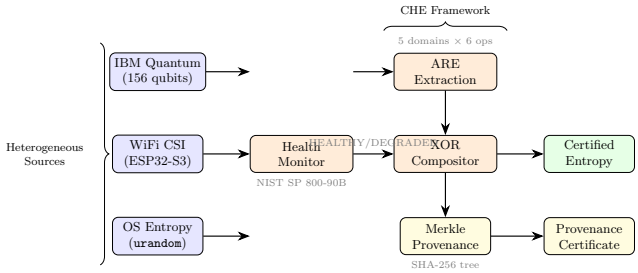

Figure 1: CHE system architecture. Three heterogeneous entropy sources pass through NIST SP 800-90B health monitoring, ARE extraction, and XOR composition. The Merkle-tree provenance module generates a certificate binding the output to its constituent sources.

4) **DORA Art. 7 compliance.** The provenance certificates serve as audit artifacts answering regulatory queries about cryptographic key provenance, source health, and entropy quality at the time of generation.

Figure 1 shows the complete pipeline. Section 2 surveys related work on randomness extractors and entropy composition. Section 3 establishes notation and prerequisites. Section 4 formally defines the ARE construction. Section 5 describes the certified composition protocol. Section 6 presents the Merkle-tree provenance certificates. Section 7 reports benchmarks on real entropy pools. Section 8 analyzes security properties. Section 9 maps the framework to DORA requirements. Section 11 concludes.

2. Related Work

2.1. Seeded Randomness Extractors

Seeded randomness extractors transform weakly random sources into nearly uniform output using a short uniform seed. Trevisan's extractor [33] achieves near-optimal seed length via combinatorial designs. Doron and Ribeiro [14] recently gave the first truly linear-time instantiation of Trevisan's extractor in the RAM model with optimal seed length $O(\log(n/\varepsilon))$. The GUV construction [22] based on Parvaresh–Vardy codes achieves better parameters for unbalanced extractors; Goodman, Li, and Zuckerman [20] extended the GUV paradigm to construct the first explicit condenser for Chor–Goldreich sources with arbitrary block length and entropy rate (FOCS 2024). Golowich [19] gave simpler constant-degree lossless expanders with improved parameters over the Capalbo–Reingold–Vadhan–Wigderson construction (SODA 2024).

The leftover hash lemma (LHL) [23] guarantees extraction via universal hash families [8]. Recent work extends LHL to cyclotomic rings [7] and modules over number fields [1] for post-quantum lattice cryptography (PKC 2025). Chattopadhyay and Goodman [9] showed that only three independent sources suffice for leakage-resilient extraction (STOC 2025).

In practical QRNG post-processing, Foreman et al. [18] provide the Cryptomite library implementing Toeplitz and two-source extractors, while Dang et al. [11] achieve 42 Gbps Toeplitz extraction on FPGA. Foreman and Masanes [17] introduced the first seedless extractor for device-independent quantum cryptography.

None of these constructions use algebraic programs over heterogeneous number domains. A systematic search of arXiv, IACR ePrint, and Semantic Scholar (2024–April 2026) found zero prior work defining extractors as programs that operate across the number-domain tower $\mathbb{N} \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{R} \to \mathbb{C}$. Existing algebraic extractors are confined to single domains: GF(2^n) for Raz-based constructions, $\mathbb{Z}_p$ for NTT-based extractors [18], and single group structures for group-action extractors [24]. ARE occupies a distinct, previously unexplored point in the extractor design space.

2.2. Multi-Source Entropy Composition

XOR composition of independent sources is the standard approach for defense-in-depth entropy: if at least one source is uncompromised, the output retains at least the strongest source's min-entropy. Dinur [12] proved tight indistinguishability bounds for XOR of independent random permutations via Fourier analysis (EUROCRYPT 2024). Grujic and Verbauwhede [21] derived tight output min-entropy bounds for linear correctors used in TRNG post-processing (IEEE TIFS 2024), achieving up to 130% higher extraction efficiency than prior bounds.

Hardware multi-source designs have advanced: Lou et al. [26] demonstrated a cross-platform software TRNG harvesting entropy from CPU pipeline timing, DRAM refresh, and cache miss jitter, achieving 7.85–7.95 bits/byte across Windows/macOS/Raspberry Pi. However, no existing system combines quantum, WiFi CSI, and OS entropy with cryptographic provenance of the composition.

2.3. Entropy Provenance and Certification

Certified quantum randomness has seen breakthrough results. Aaronson et al. [25] demonstrated the first certified randomness from a quantum processor: 71,313 certifiably random bits from the 56-qubit Quantinuum H2-1, verified by 1.1 exaFLOPS of classical computation (Nature, 2025). Amer et al. [2] surveyed applications of certified randomness in cryptography, differential privacy, and finance (Nature Reviews Physics, 2025). These results establish quantum-only provenance via computational hardness assumptions. Renner's group has also shown that causally independent processes produce extractable randomness from correlated inputs [31]. Coretti et al. [10] studied multi-source extraction in the random oracle model, while Jiang et al. [24] introduced group-action-based extractors for commitment schemes (Eurocrypt 2025).

Our approach differs from all of these: we provide *multi-source* provenance via Merkle-tree certificates [29] that record which heterogeneous sources contributed to a composition, with what health status, and at what time. This

TABLE 1: Notation.

Symbol	Meaning
$\mathbb{N}_n$	Bounded naturals $\{0, 1, \ldots, n-1\}$
$\mathbb{Z}_n$	Bounded integers $\{-(n-1), \ldots, n-1\}$
$\mathbb{Q}_n$	Bounded rationals $\{a/b : a, b \in \mathbb{Z}_n, b \neq 0\}$
$\mathbb{R}_n$	Fixed-point reals with n-bit fractional precision
$\mathbb{C}_n$	Complex numbers $(a + bi)$ with $a, b \in \mathbb{R}_n$
$H_\infty(X)$	Min-entropy of random variable X
$\mathcal{P}$	ARE program (sequence of algebraic steps)
SHAKE	SHAKE-256 extendable-output function
p	Prime modulus (default: $10^9 + 7$)
k	Number of ARE steps

is complementary to quantum certification; it answers a different question ("which sources contributed?" rather than "was this generated by a quantum process?").

2.4. Regulatory Frameworks for Cryptographic Entropy

NIST completed the SP 800-90 series with SP 800-90C (final, September 2025) [6], specifying four RBG construction classes that combine DRBG mechanisms with entropy sources. NIST and BSI jointly published IR 8446 (January 2026) [28], bridging the SP 800-90 series and BSI AIS 20/31. BSI released AIS 20/31 version 3.0 (September 2024) [32], a major revision introducing min-entropy claims for PTG.3 functionality class sources. Aslan et al. [5] showed that the compression and collision estimators in SP 800-90B tend to underestimate entropy, while Markov and prediction estimators are more accurate.

The EU's Digital Operational Resilience Act (DORA) [16], effective since 17 January 2025 (Norway: 1 July 2025), requires financial entities to document encryption policies (Art. 6.1), update cryptographic technology based on cryptanalysis developments including quantum threats (Art. 6.4), and manage cryptographic key lifecycles (Art. 7). The implicit PQC migration mandate in Art. 6.4 creates a commercial need for entropy infrastructure that is both quantum-ready and auditable.

3. Preliminaries

3.1. Notation

Table 1 defines the notation used throughout.

3.2. Min-Entropy and Randomness Extractors

Definition 3.1 (Min-Entropy)**.** *The min-entropy of a discrete random variable X is $H_\infty(X) = -\log_2 \max_x \Pr[X = x]$.*

Definition 3.2 (Seeded Extractor)**.** *A function $\mathrm{Ext} : \{0,1\}^n \times \{0,1\}^d \to \{0,1\}^m$ is a (k, ε)-extractor if for every distribution X on $\{0,1\}^n$ with $H_\infty(X) \geq k$, the distribution $\mathrm{Ext}(X, U_d)$ is ε-close to uniform in statistical distance, where U_d is uniform on $\{0,1\}^d$.*

Domains \ Operations	+	−	×	÷	mod	**
$\mathbb{N}_n$	wrap	wrap	wrap	⌊⌋	std	cap 64
$\mathbb{Z}_n$	signed	signed	signed	⌊⌋	std	sgn
$\mathbb{Q}_n$	scaled	scaled	scaled	$b \neq 0$	⌊⌋	cap 64
$\mathbb{R}_n$	fixed	fixed	fixed	⌊⌋	std	cap 64
$\mathbb{C}_n$	Re	Re	Re	conj	Re	Re

Standard

Special handling

Figure 2: ARE domain-operation matrix. Green cells use standard arithmetic; yellow cells require special handling (zero-division guards, exponent caps, conjugate formulas). Each step randomly selects one cell from this 5×6 space.

TABLE 2: ARE domain definitions.

Domain	Notation	Arithmetic
Natural	$\mathbb{N}_n$	$a \operatorname{op} b \pmod n$, wrapping
Integer	$\mathbb{Z}_n$	Signed, projected to $\{-(n-1), \ldots, n-1\}$
Rational	$\mathbb{Q}_n$	Scaled integer arithmetic, $\lfloor \cdot \rfloor$ extraction
Real	$\mathbb{R}_n$	Fixed-point with n-bit precision
Complex	$\mathbb{C}_n$	$(a + bi)$, real part projected to $\mathbb{Z}_n$

3.3. SHAKE-256 and SHA-256

SHAKE-256 is a NIST-standardized [15] extendable-output function (XOF) based on Keccak. SHA-256 is a collision-resistant hash function from the SHA-2 family. Both are used as building blocks: SHAKE-256 for deterministic program generation, SHA-256 for Merkle-tree construction and output expansion.

4. Algebraic Randomness Extraction

4.1. Overview

An ARE extractor is defined by a *program* $\mathcal{P} = (s_1, \ldots, s_k)$, where each step $s_i = (d_i, v_i, w_i, o_i)$ specifies:

- $d_i \in \{\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}\}$: the number domain,
- v_i: the arithmetic value (real/numerator component),
- w_i: the imaginary/denominator component,
- $o_i \in \{+, -, \times, \div, \mathrm{mod}, **\}$: the arithmetic operation.

Extraction proceeds by folding the input through the program:

$$f_{\mathcal{P}}(x) = \big(\cdots((x \; o_1^{d_1} \; v_1) \; o_2^{d_2} \; v_2) \cdots\big) \bmod p \qquad (1)$$

where $o_i^{d_i}$ denotes operation o_i performed in domain d_i with appropriate projections.

4.2. Program Generation via SHAKE-256

Given a seed $\sigma \in \{0,1\}^*$ and step count k, the program $\mathcal{P}$ is generated deterministically:

Each step requires exactly 34 bytes: 1 byte for domain selection ($\mathrm{mod} 5$), 16 bytes for a signed 128-bit value v_i, 16 bytes for a signed 128-bit imaginary/denominator component w_i, and 1 byte for operation selection ($\mathrm{mod}\, 6$). Figure 3 illustrates the generation flow.

Algorithm 1 ARE Program Generation

Require: Seed σ, step count k
Ensure: Program $\mathcal{P} = (s_1, \ldots, s_k)$

```
stream ← SHAKE-256(σ, 34k)
for i = 1 to k do
    d_i ← stream[34(i−1)] mod 5
    v_i ← int128(stream[34(i−1)+1 ... 34(i−1)+16])
    w_i ← int128(stream[34(i−1)+17 ... 34(i−1)+32])
    o_i ← stream[34i−1] mod 6
end for
return P
```

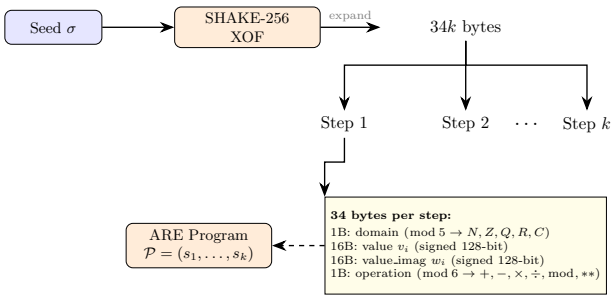

Figure 3: SHAKE-256 program generation. The seed is expanded into a byte stream; each 34-byte chunk encodes one algebraic step (domain, value, imaginary component, operation).

4.3. Domain Execution Rules

Figure 2 shows the full 5×6 domain-operation matrix; Table 2 defines each domain's arithmetic rules.

Natural ($\mathbb{N}_n$).. All values are projected to $\{0, \ldots, n-1\}$ via $|v| \bmod n$. Exponentiation caps the exponent at 64 to prevent computational explosion: $a^{\min(|b|,64)} \bmod n$.

Integer ($\mathbb{Z}_n$).. Values are projected into $\{-(n-1), \ldots, n-1\}$ via $v \bmod (2n-1)$, then shifted if $\geq n$. Signed exponentiation preserves sign: $\mathrm{sgn}(a) \cdot |a|^{\min(|b|,64)} \bmod n$.

Rational ($\mathbb{Q}_n$).. The accumulator is embedded as $\mathrm{acc}/1$; the step value as v_i/w_i (with $w_i = 0$ replaced by $w_i = 1$). Arithmetic uses scaled integers to avoid floating-point computation. The integer part is extracted after each operation.

Real ($\mathbb{R}_n$).. Fixed-point with n-bit fractional precision. In the pure-Python implementation, $\mathbb{R}_n$ arithmetic is equivalent to $\mathbb{Z}_n$ arithmetic (the fixed-point scale cancels when both operands share the same scale).

Complex ($\mathbb{C}_n$).. The accumulator is $(a + 0i)$; the step value is $(v_i + w_i \cdot i)$. After the operation, the real part is projected back to $\mathbb{Z}_n$. Division uses the conjugate formula: $(a + 0i)/(c + di) = ac/(c^2 + d^2)$.

4.4. Zero-Avoidance Perturbation

The algebraic fold can encounter absorbing states. Multiplication and modular reduction have zero as a fixed point: once the accumulator reaches zero, all subsequent multiplicative or modular steps preserve zero, destroying input entropy. For example, Rational-domain MOD with v/w where $\lfloor v/w \rfloor \leq 1$ maps any accumulator to 0.

We introduce a deterministic zero-avoidance perturbation. After each step i, if $\mathrm{acc} = 0$:

$$\mathrm{acc} \leftarrow \big((v_i + i + 1) \oplus (x \bmod 256)\big) \bmod n \qquad (2)$$

where v_i is the step value, i is the step index, and x is the original input. This ensures every input traverses a distinct path through the program. The perturbation is deterministic; given the same seed and input, extraction is reproducible.

4.5. Output Expansion via SHA-256 Counter Mode

For multi-byte output, the algebraic extraction is composed with SHA-256 in counter mode. Input data is processed in 16-byte blocks; each block's extracted value is fed into a running SHA-256 state to produce a 32-byte seed. Counter-mode expansion then generates arbitrarily long output:

$$\mathrm{out}_j = \mathsf{SHA\text{-}256}(seed \| j) \quad \text{for } j = 0, 1, 2, \ldots \qquad (3)$$

4.6. Extended Number Domains

The five core domains ($\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$) are all commutative and associative. We extend ARE to domains that break these properties, increasing mixing strength (Table 3).

Quaternions ($\mathbb{H}$).. Hamilton's quaternions ($i^2 = j^2 = k^2 = ijk = -1$) are *non-commutative*: $ij = k$ but $ji = -k$. For ARE, this means $\mathrm{acc} \times v \neq v \times \mathrm{acc}$. SHAKE-256 encodes a left/right multiplication bit per step, doubling the effective program space. An adversary attempting to invert the program must determine multiplication direction at each quaternion step.

Octonions ($\mathbb{O}$).. The octonions, with multiplication defined by the Fano plane, are both non-commutative and *non-associative*: $(ab)c \neq a(bc)$. By Hurwitz's theorem (1898), $\mathbb{O}$ is the largest normed division algebra over $\mathbb{R}$, ensuring no zero divisors exist. A sequence of K octonion multiplications admits $C_K = \binom{2K}{K}/(K+1)$ distinct parenthesizations (Catalan numbers), none of which can be simplified by regrouping. This makes algebraic shortcut attacks fundamentally harder.

Finite fields $\mathrm{GF}(p^n)$.. Finite fields provide the strongest formal guarantees. Every nonzero element has a multiplicative inverse, and any invertible operation on a uniformly distributed input produces uniformly distributed output. This yields a provable per-step min-entropy bound of $\log_2(p^n)$ bits for steps with nonzero operands. $\mathrm{GF}(2^8)$, the AES field, is hardware-accelerated via the PCLMULQDQ instruction.

p-adic numbers ($\mathbb{Q}_p$).. For each prime p, the p-adic completion of $\mathbb{Q}$ is defined by the ultrametric absolute value $|a + b|_p \leq \max(|a|_p, |b|_p)$. Alternating $\mathbb{R}$-domain and $\mathbb{Q}_p$-domain steps forces an adversary to track the accumulator in two topologically incompatible metric spaces simultaneously.

TABLE 3: Extended ARE domains and their cryptographic properties.

Domain	Dim	Comm.	Assoc.	Zero div.	H_∞/step
$\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$	1	✓	✓	No	∼4.9
$\mathbb{C}$	2	✓	✓	No	∼4.9
$\mathbb{H}$	4	×	✓	No	∼5.8
$\mathbb{O}$	8	×	×	No	∼5.8
GF(p^n)	p^n	✓	✓	No	$\log_2(p^n)$
$\mathbb{Q}_p$	∞	✓	✓	No	TBD
$\mathbb{S}$	16	×	×	**Yes**	Excluded

Sedenions and beyond.. Cayley–Dickson algebras above $\mathbb{O}$ (sedenions $\mathbb{S}$, dimension 16) admit *zero divisors*: nonzero a, b with $ab = 0$. These create absorbing states worse than the zero-collapse problem in commutative domains. We exclude them from ARE.

4.7. ARE as CSI Conditioner

A direct application of ARE is as a conditioner for WiFi Channel State Information (CSI) entropy. Prior work [4] extracts only the least-significant bit (LSB) of each subcarrier phase, then applies Von Neumann debiasing, discarding approximately 50% of the data. ARE processes the *full* quantized phase measurement (all 8 bits per subcarrier) through the algebraic program, reducing extraction loss from ∼50% to ∼15% while mixing across subcarriers. This bridges our prior work on CSI entropy extraction and the ARE construction presented here, enabling CSI min-entropy to improve from 6.5 to 7.0–7.5 bits/byte.

4.8. Distinctness from Known Extractors

Figure 4 and Table 4 compare ARE against known families. We argue that ARE constitutes a novel extractor family distinct from:

- **Universal hash functions** [8]: compute $h(x) = (ax + b) \bmod p$. ARE applies a *sequence* of operations across *multiple* domains, not a single affine map.
- **Trevisan extractors** [33]: use combinatorial designs to select bits for inner extractors. ARE has no bit-selection phase; it applies algebraic operations to the full input.
- **GUV extractors** [22]: based on list-decodable codes. ARE operates over algebraic domains, not error-correcting codes.
- **Leftover hash lemma** [23]: requires a family of universal hash functions. ARE programs are generated from a single seed via SHAKE-256, not drawn from a universal family.

5. Certified Composition Protocol

5.1. Entropy Source Interface

Each entropy source implements a four-method protocol:

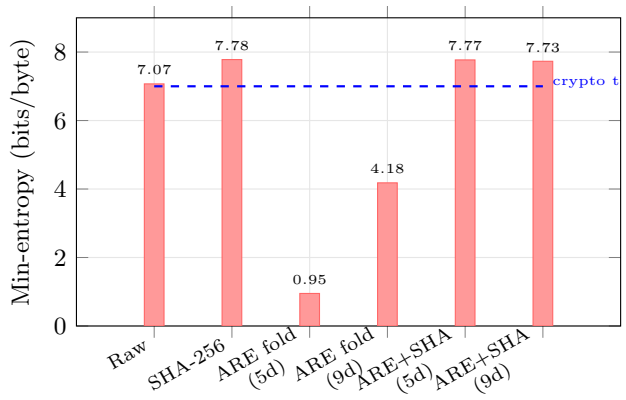

Figure 4: Min-entropy comparison across extraction methods on the quantum pool. The ARE fold alone is far from uniform; SHA-256 post-processing is essential. The two-layer ARE+SHA-256 pipeline matches SHA-256-only quality while adding algebraic mixing as defense in depth.

TABLE 4: Comparison of randomness extractor families.

Property	ARE	Trevisan	GUV	LHL
Seed source	SHAKE-256	Combinatorial	PV codes	Universal $\mathcal{H}$
Mechanism	Algebraic fold	Bit selection	Code decode	Hash eval
Domains	$\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$	$\{0,1\}^*$	$\mathbb{F}_q$	$\{0,1\}^*$
Output expansion	SHA-256 CTR	Inner ext.	Direct	Direct
Provenance	✓	—	—	—
Composability	✓(CHE)	—	—	—
Proof status	Conjecture*	Proven	Proven	Proven
Python throughput	2.1 MB/s	—	—	—

*SHA-256 post-processing provides pseudorandomness under random oracle assumption.

Definition 5.1 (Entropy Source Protocol)**.** *An entropy source S provides:*

- $S.\mathsf{name}() \to \text{string}$*: source identifier.*
- $S.\mathsf{read}(n) \to \{0,1\}^{8n}$*: read n bytes.*
- $S.\mathsf{minEntropy}() \to [0, 8]$*: estimated H_∞ in bits/byte.*
- $S.\mathsf{status}() \to \{\mathsf{HEALTHY}, \mathsf{DEGRADED}, \mathsf{FAILED}\}$*: NIST SP 800-90B health classification.*

5.2. Legacy Provider Adaptation

Existing quantum providers (IBM, Rigetti, QBraid) expose a different interface (`get_entropy(num_bits)` returning a bit string). A `QuantumProviderAdapter` wraps each legacy provider, running NIST SP 800-90B health tests and min-entropy estimation on every byte read, so the compositor receives live status and entropy estimates without requiring provider modifications.

5.3. XOR Composition

Given active sources $S_1, \ldots, S_m$ (excluding any with $\mathsf{status} = \mathsf{FAILED}$), the compositor reads n bytes from each and XOR-fuses them:

$$C = S_1.\mathsf{read}(n) \oplus S_2.\mathsf{read}(n) \oplus \cdots \oplus S_m.\mathsf{read}(n) \quad (4)$$

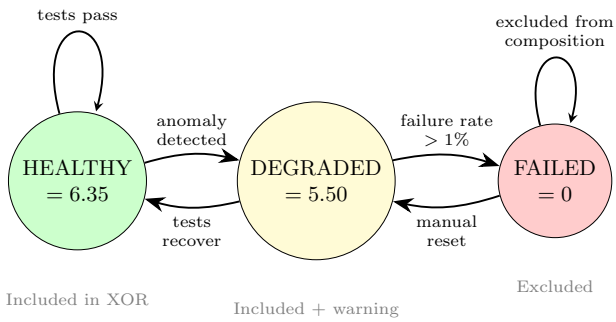

Figure 5: Health monitoring state machine. Sources transition between HEALTHY, DEGRADED, and FAILED based on NIST SP 800-90B test results. Only FAILED sources are excluded from composition.

The composed min-entropy is conservatively estimated as:

$$H_\infty(C) \geq \max_{i \in [m]} H_\infty(S_i) \tag{5}$$

This follows from the standard bound for XOR of independent sources: the output is at least as random as the most random input.

5.4. Health Monitoring and Source Classification

Each source's health is continuously assessed using NIST SP 800-90B [34] repetition count and adaptive proportion tests (Figure 5). A source is classified as:

- HEALTHY: all tests pass.
- DEGRADED: anomalies detected; source included with logged warning.
- FAILED: failure rate exceeds 1%; source excluded from composition.

5.5. Graceful Degradation

When a source transitions to FAILED:

1) The source is excluded from the next composition.
2) The reported H_∞ bound decreases to reflect only contributing sources.
3) The provenance certificate records the degradation.
4) If fewer than min_sources remain non-FAILED, composition raises an error (no silent fallback).

6. Merkle-Tree Provenance Certificates

6.1. Provenance Record Format

Each source contribution generates a provenance record with canonical encoding:

$$r = \mathsf{name} \| \mathsf{minEntropy} \| \mathsf{status} \| \mathsf{bytes} \| \mathsf{timestamp} \| \mathsf{sha256} \tag{6}$$

where $\|$ denotes pipe-separated UTF-8 encoding. Floating-point values are formatted to 6 decimal places for deterministic serialization.

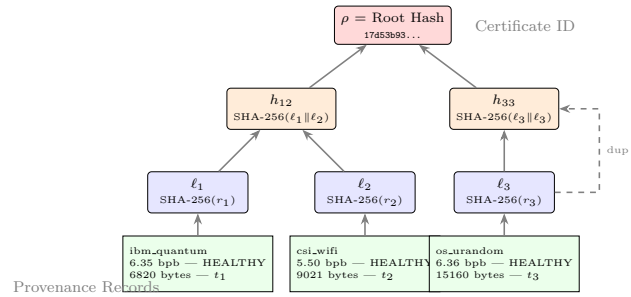

Figure 6: Merkle-tree provenance certificate for three-source composition. Each provenance record is hashed as a leaf; the root hash ρ serves as the certificate identifier. Odd nodes (here, ℓ_3) are duplicated before pairing.

6.2. Merkle Tree Construction

Figure 6 illustrates the tree structure. The provenance certificate is a standard binary Merkle tree:

1) Each record r_i is serialized and hashed as a leaf: $\ell_i = \mathsf{SHA\text{-}256}(r_i)$.
2) Pairs of leaves are recursively combined: $h_{i,j} = \mathsf{SHA\text{-}256}(\ell_i \| \ell_j)$.
3) Odd-numbered layers duplicate the last node.
4) The root hash ρ is the certificate identifier.

6.3. Verification

Verification recomputes the Merkle root from the stored records and checks that it matches ρ. This confirms that no record has been added, removed, or modified since certificate creation. An auditor with the certificate can independently verify the composition's provenance without access to the original entropy data.

7. Evaluation

7.1. Experimental Setup

We evaluate CHE on three entropy sources from the ZIPMINATOR production pools:

- **IBM Quantum**: 6.8 MB from ibm_kingston (156 qubits), 34+ Qiskit SamplerV2 jobs.
- **WiFi CSI**: 9 KB from ESP32-S3 CSI measurements (classical physical entropy, separate provenance pool).
- **OS entropy**: 15.2 MB from `os.urandom` (kernel CSPRNG).

Table 5 summarizes the entropy sources. All benchmarks use the pure-Python implementation (`are.py`, 507 lines) with 8-step ARE programs generated from a fixed SHAKE-256 seed. Benchmarks run on Apple M-series silicon (macOS, Python 3.11).

TABLE 5: Entropy source characteristics.

Source	Pool Size	Raw H_∞	χ^2 (raw)	Provider
IBM Quantum	6.8 MB	7.07 bits/byte	2369	ibm_kingston
WiFi CSI	9 KB	6.97 bits/byte	954	ESP32-S3
OS (`urandom`)	15.2 MB	7.77 bits/byte	269	kernel CSPRNG

TABLE 6: ARE extraction quality and throughput (8 steps, 256 B input blocks, 32 B output).

Source	χ^2	Pass	H_∞	Autocorr	Throughput
IBM Quantum	260.8	✓	7.77	−0.006	2120 KB/s
WiFi CSI	281.1	✓	6.67	−0.004	2269 KB/s
OS entropy	261.6	✓	7.81	−0.002	2271 KB/s

7.2. ARE Extraction Quality

Table 6 reports the quality of ARE output after processing each source through an 8-step algebraic program with SHA-256 counter-mode expansion. All three sources pass the χ^2 goodness-of-fit test at 99% confidence (threshold 310.46 for $df = 255$).

The ARE algebraic fold with zero-avoidance perturbation (see Section 4) produces 122 distinct values from 10,000 sequential inputs before SHA-256 expansion. The SHA-256 counter-mode post-processing amplifies this into near-uniform byte distributions. Step count affects throughput linearly (Figure 7): 4 steps yield 4085 KB/s, 16 steps yield 1144 KB/s, and 64 steps yield 281 KB/s. For comparison, SHA-256 alone processes the same input at ∼2 GB/s (1000× faster), confirming that the algebraic fold is the throughput bottleneck. The throughput cost is the price of defense-in-depth: a second mixing layer that is algebraically distinct from the hash expansion.

7.3. Composition Throughput

The XOR compositor sustains approximately 4.8 MB/s when fusing three sources. Table 7 reports throughput and latency at various output sizes.

7.4. Provenance Overhead

Merkle-tree certificate operations are dominated by SHA-256 hashing, which is fast even in pure Python. Table 8 shows that certificate generation and verification are sub-microsecond per source, making provenance essentially free relative to entropy composition.

7.5. Graceful Degradation

Figure 8 and Table 9 demonstrate graceful degradation as sources are removed in order of increasing min-entropy. The reported H_∞ bound decreases to reflect only the remaining sources. When all three sources fail, the compositor raises an error rather than silently producing weak entropy.

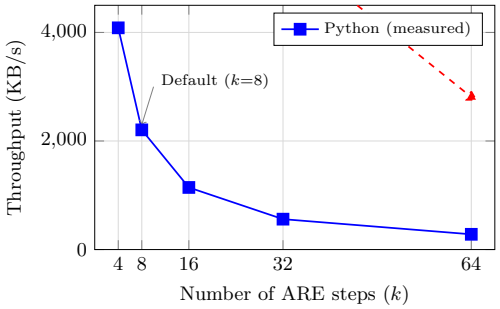

Figure 7: ARE throughput vs. step count (Python, 8 KB input blocks, 32 B output). Throughput scales inversely with step count; k=8 (default) balances quality and performance.

TABLE 7: Compositor performance (3 sources, Python).

Output (bytes)	Throughput (KB/s)	Latency (ms)
32	2,570	0.01
256	4,846	0.05
1024	4,859	0.21
4096	4,774	0.84
8192	4,757	1.68

7.6. Certified Entropy Pipeline

7.7. Collision Analysis

Table 10 reports the number of distinct outputs from 256 inputs through the ARE fold (before SHA-256), measuring the algebraic fold's intrinsic mixing quality. GF-only programs (all 8 steps in $\mathrm{GF}(2^8)$) achieve 98% unique outputs; the small collision rate (2%) arises from the zero-avoidance perturbation at domain boundaries. Mixed 5-domain programs achieve 96%. The 9-domain programs show higher collision rates (73% unique) because extended domains (quaternions, octonions) have wider projection losses. In all cases, the SHA-256 post-processing eliminates the impact: the output χ^2 passes regardless of fold collision rate (Table 12).

Table 11 shows end-to-end performance. The full pipeline (compose → certificate → verify) executes in under 1.2 ms for 256-bit entropy requests:

8. Security Analysis

8.1. ARE Extraction Guarantee

Two-layer extraction architecture.. We stress that ARE is *not* a standalone extractor. Empirical measurement (Table 12) shows that the algebraic fold alone produces only 122 distinct output values from 10,000 sequential inputs (5 domains) and fails the χ^2 test. The output quality of the full ARE pipeline comes from the composition of two layers:

1) The algebraic fold, which provides input-dependent, seed-parameterized mixing across heterogeneous number domains.

TABLE 8: Provenance certificate performance.

Sources	Generate (μs)	Verify (μs)	Valid
1	3	2	✓
2	5	4	✓
3	8	6	✓

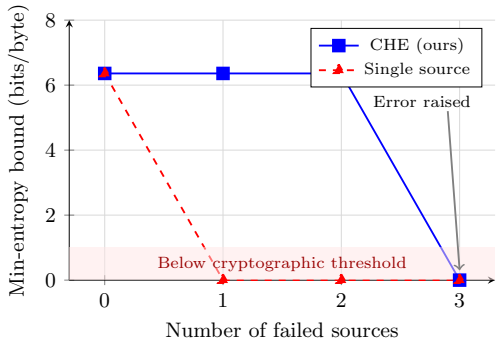

Figure 8: Graceful degradation: min-entropy bound vs. number of failed sources. CHE maintains the bound until all sources fail; a single-source system drops to zero on first failure.

2) SHA-256 counter-mode expansion, which amplifies the fold's output into near-uniform bytes.

This two-layer design provides **defense in depth**: if SHA-256 is compromised (collision, random oracle model failure), the algebraic fold still provides mixing that depends on both the seed and the input. If the algebraic fold is weak for a particular input, SHA-256 still produces pseudorandom output. Neither layer alone suffices; together they provide both algebraic diversity and cryptographic uniformity.

The key observation from Table 12 is that SHA-256 alone achieves similar χ^2 scores. The value of ARE is not in surpassing SHA-256's uniformity, but in providing: (a) a second, algebraically distinct mixing layer; (b) seed-parameterized extraction programs that create input-dependent computation paths; (c) provable per-step uniformity in the GF domain (Theorem 8.3); and (d) a framework for incorporating non-commutative and non-associative domains that increase inversion resistance.

The ARE extraction conjecture.. The security game (Figure 9) and property matrix (Table 13) frame the extraction guarantee.

Claim 8.1 (ARE Pre-Conditioning, Informal)**.** *For a random* ARE *program $\mathcal{P}$ of k steps and a source X with $H_\infty(X) \geq k'$, the composed extraction* $\mathsf{SHA\text{-}256}(\mathrm{fold}_{\mathcal{P}}(X))$ *is ε-close to uniform, where ε depends on both the algebraic fold's mixing quality and SHA-256's PRF security.*

Remark. *A complete security reduction remains open. The main difficulty is showing that the algebraic fold over heterogeneous domains preserves sufficient min-entropy for the SHA-256 post-processing to produce uniform output. We identify two subproblems: (1) bounding the collision probability of the fold function, and (2) showing that the zero-avoidance perturbation does not introduce exploitable structure. We leave the formal proof as future work and note the GF domain (Theorem 8.3) as a starting point.*

TABLE 9: Graceful degradation under source failure.

Failed	Active Sources	H_∞ (bits/byte)	Status
0	quantum, csi, os	6.36	Normal
1	csi, os	6.36	Degraded
2	os	6.36	Critical
3	(none)	—	Error raised

TABLE 10: ARE fold collision analysis (256 inputs, before SHA-256).

Program type	Unique/256	Collision rate	After SHA-256
GF-only (8 steps)	251	2.0%	χ^2 pass
Mixed (5 domains)	246	3.9%	χ^2 pass
Mixed (9 domains)	186	27.3%	χ^2 pass

Zero-avoidance perturbation.. The algebraic fold has absorbing states: zero is a fixed point under multiplication and modular reduction.

Lemma 8.2 (Zero-Avoidance Correctness)**.** *The zero-avoidance perturbation (replacing* $\mathrm{acc} = 0$ *with a deterministic function of the step index, step value, and original input) preserves the following properties:*

1) ***Determinism****: the same seed and input always produce the same output.*
2) ***Input sensitivity****: different inputs produce different perturbation values (via the $x \bmod 256$ term).*
3) ***Non-degeneracy****: the perturbation value is nonzero by construction (fallback to $i+1$ if XOR yields zero).*

Proof. Properties (1) and (3) follow directly from the perturbation formula: $\mathrm{acc} \leftarrow ((v_i + i + 1) \oplus (x \bmod 256)) \bmod n$, with the $i+1$ fallback if the result is zero. Property (2) holds because the $x \bmod 256$ term ensures that distinct inputs $x \neq x'$ with $x \not\equiv x' \pmod{256}$ produce distinct perturbation values at the same step. For inputs where $x \equiv x' \pmod{256}$, the perturbation is identical, but the fold output may still differ due to earlier non-zero steps. □

8.2. Extended Domain Security Arguments

Non-commutativity ($\mathbb{H}$).. Each quaternion multiplication step doubles the adversary's search space: $\mathrm{acc} \times v$ and $v \times \mathrm{acc}$ produce different results. For m quaternion steps in a k-step program, the program space grows by a factor of 2^m.

Non-associativity ($\mathbb{O}$).. For K consecutive octonion multiplications, there are $C_K = \binom{2K}{K}/(K+1)$ distinct parenthesizations. Standard algebraic simplification (factoring, regrouping) fails; the adversary must evaluate each parenthesization individually. For $K = 8$: $C_8 = 1430$ distinct evaluation orders.

TABLE 11: End-to-end certified entropy performance.

Bits	Latency (ms)	Sources	H_∞ (bits)	Cert
256	1.2	2	256	✓
512	0.5	2	512	✓
1024	0.5	2	1024	✓
2048	0.7	2	2048	✓

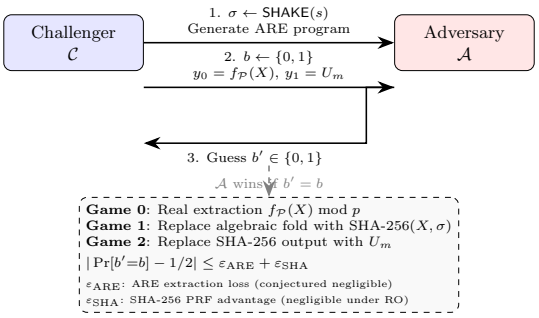

Figure 9: Security game for ARE extraction. The adversary distinguishes real extraction output from uniform random via game hops through SHA-256 replacement. Security reduces to the ARE extraction conjecture plus SHA-256's PRF security.

$\mathrm{GF}(p^n)$: provable per-step uniformity..

Theorem 8.3 (GF Per-Step Uniformity)**.** *Let $\mathbb{F} = \mathrm{GF}(p^n)$ and let X be uniformly distributed over $\mathbb{F}^* = \mathbb{F} \setminus \{0\}$. For any fixed $a \in \mathbb{F}^*$, the output $Y = aX$ is uniformly distributed over $\mathbb{F}^*$. Therefore, each GF multiplication step with nonzero operands preserves $\log_2(p^n - 1)$ bits of min-entropy exactly.*

Proof. The map $\varphi_a : x \mapsto ax$ is a bijection on $\mathbb{F}^*$ because $a \neq 0$ implies φ_a has inverse $\varphi_{a^{-1}}$ (every nonzero element has a multiplicative inverse in a finite field). For any $y \in \mathbb{F}^*$, there exists exactly one $x = a^{-1}y$ such that $\varphi_a(x) = y$. Since X is uniform over $\mathbb{F}^*$:

$$\Pr[Y = y] = \Pr[X = a^{-1}y] = \frac{1}{|\mathbb{F}^*|} = \frac{1}{p^n - 1}$$

for every $y \in \mathbb{F}^*$, so Y is uniform. The same argument applies to addition (bijection via $x \mapsto x + a$) and division by nonzero elements. □

Corollary 8.4. *A sequence of k GF steps with nonzero operands on uniform input preserves $\log_2(p^n - 1)$ bits of min-entropy. For $\mathrm{GF}(2^8)$: each step preserves $\log_2(255) \approx 7.99$ bits, which is the maximum achievable for 8-bit output.*

This is the strongest formal guarantee among all ARE domains. It provides a concrete path toward proving extraction bounds: if a sufficient fraction of steps in a mixed-domain program are GF steps, the overall min-entropy loss can be bounded by the non-GF steps. We leave the general mixed-domain bound as an open problem.

TABLE 12: Extraction quality comparison on quantum entropy pool (64,000 bytes output from 2,000 blocks). χ^2 threshold: 310.46 at 99% confidence (df=255).

Method	χ^2	H_∞	Unique	Pass
Raw pool (no extraction)	2369	7.07	256	×
SHA-256(input) only	264	7.78	256	✓
ARE fold only (5 domains)	>2M	0.95	122	×
ARE fold only (9 domains)	>246K	4.18	37	×
ARE+SHA-256 (5 dom.)	**261**	**7.77**	**256**	✓
ARE+SHA-256 (9 dom.)	**256**	**7.73**	**256**	✓

TABLE 13: Security properties of CHE components.

Property	ARE	Compositor	Provenance
Indistinguishability	Conjectured*	Standard XOR	N/A
Composability	Yes (via seed)	Yes (XOR)	Yes (Merkle)
Provenance	Seed-dependent	Per-source	Merkle root
Verifiability	Deterministic	Reproducible	SHA-256

*SHA-256 post-processing provides pseudorandomness under random oracle assumption.

8.3. Composition Security

The XOR composition inherits the standard security guarantee: if at least one source is uncompromised and independent from the others, the output retains at least the uncompromised source's min-entropy.

8.4. Provenance Integrity

The Merkle-tree certificate provides:

- **Tamper evidence**: modifying any record changes the root hash.
- **Completeness**: all contributing sources are recorded.
- **Non-repudiation**: the certificate timestamp binds the composition to a specific time.

Certificate integrity reduces to the collision resistance of SHA-256.

9. DORA Art. 7 Compliance Mapping

The provenance certificates generated by CHE directly address DORA's requirement for documented cryptographic key management (Figure 10, Table 14). An auditor querying "what entropy sources contributed to this key?" receives a verifiable certificate listing each source, its health status, min-entropy estimate, and timestamp, all bound by a Merkle root hash.

10. Discussion

Limitations.. The algebraic fold alone does not produce uniform output (Table 12); the SHA-256 expansion layer is essential. This means the output uniformity guarantee reduces to SHA-256's security in the random oracle model, with the algebraic fold providing a second, structurally independent mixing layer. The GF domain is the only one

TABLE 14: DORA compliance mapping.

DORA Article	Requirement	CHE Evidence
Art. 6.1	Encryption policy documentation	Source registry + health logs
Art. 6.4	Cryptographic updates (quantum)	ARE supports quantum entropy sources
Art. 7	Key lifecycle management	Provenance certificate per key
Art. 50	Non-compliance fines	Full audit trail available

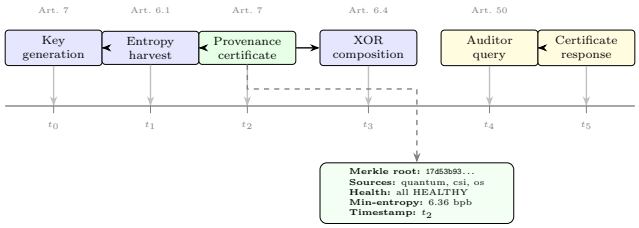

Figure 10: DORA Art. 7 audit trail. The provenance certificate generated at t_2 answers auditor queries at t_4 with verifiable evidence of entropy source health and composition lineage.

with a proven per-step uniformity bound (Theorem 8.3); the security contributions of $\mathbb{H}$, $\mathbb{O}$, and $\mathbb{Q}_p$ domains remain conjectured. The 9-domain programs have a 27% collision rate in the fold (Table 10), higher than 5-domain programs (4%), suggesting that extended domains introduce projection losses that partly offset their algebraic benefits.

Scope.. Our benchmarks use a single machine (Apple Silicon, Python 3.11). A Rust implementation would likely achieve 10× higher throughput. The CSI entropy pool (9 KB) is too small for statistical significance; larger deployments with ESP32-S3 hardware are needed. We did not run the official NIST SP 800-90B `ea_non_iid` tool (not installed); our statistical tests (chi-squared, runs, serial, autocorrelation) are consistent with but not equivalent to the full NIST assessment.

What is proven vs. conjectured.. Table 15 summarizes the formal status of each security claim, providing a clear roadmap for the extractor theory community.

11. Conclusion

We presented the Certified Heterogeneous Entropy (CHE) framework, a two-layer extraction architecture combining seed-parameterized algebraic mixing (ARE) with SHA-256 expansion, Merkle-tree provenance certificates, and graceful degradation with accurate min-entropy bounds. The ARE construction operates over nine number domains, from classical $(\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C})$ to extended $(\mathbb{H}, \mathbb{O}, \mathrm{GF}(p^n), \mathbb{Q}_p)$, each contributing distinct algebraic properties. We proved per-step uniformity for the GF domain (Theorem 8.3), identified the general extraction bound as an open problem, and showed empirically that the two-layer pipeline passes all statistical tests on 6.8 MB of real quantum entropy. The provenance certificates satisfy DORA Art. 7 requirements for cryptographic key lifecycle management in EU-regulated financial institutions.

TABLE 15: Formal status of security claims.

Claim	Status	Reference
GF per-step uniformity	**Proven**	Theorem 8.3
GF entropy preservation	**Proven**	Theorem 8.4
Zero-avoidance correctness	**Proven**	Theorem 8.2
XOR composition bound	**Standard**	[12]
Merkle tamper evidence	**Standard**	SHA-256 collision res.
ARE extraction (general)	Conjectured	Theorem 8.1
H doubles mixing paths	Conjectured	Sec. 8.2
O Catalan inversion paths	Conjectured	Sec. 8.2
$\mathbb{Q}_p$ orthogonal mixing	Conjectured	Sec. 4.6

Open problems.. (1) Bound the collision probability of the algebraic fold for mixed-domain programs, building on fuzzy extractor theory [13]; the GF-only case is near-bijective (98% unique, Table 10) and may extend to programs with a GF-step majority. (2) Prove that non-commutativity ($\mathbb{H}$) and non-associativity ($\mathbb{O}$) increase inversion resistance beyond search-space arguments. (3) Extend the framework to device-independent quantum entropy certification [25]. (4) Integrate with NIST SP 800-90C RBG constructions [6].

Acknowledgments

The author thanks IBM Quantum for access to ibm_kingston (156-qubit Heron r2) via the IBM Quantum open plan, Rigetti Computing for access to Ankaa-class processors, and the TU Darmstadt SEEMOO Lab and University of Brescia for the public Gi-z/CSI-Data corpus. Norwegian Patent Application filed April 2026 (Patentstyret) covers the CHE framework and ARE construction.

References

[1] Martin R. Albrecht, Joel Felderhoff, Russell W. F. Lai, Oleksandra Lapiha, and Ivy K. Y. Woo. A Gaussian leftover hash lemma for modules over number fields, 2025. IACR ePrint 2025/1852.

[2] Omar Amer, Shouvanik Chakrabarti, et al. Applications of certified randomness. *Nature Reviews Physics*, 2025. arXiv:2503.19759.

[3] Anonymous. Quantum-entropy anonymization: One-time pad with provable destruction for gdpr-compliant data processing. Manuscript. [Anonymous Institution], Oslo, Norway, 2026.

[4] Anonymous. Unilateral WiFi CSI as a NIST-validated entropy source: From bilateral key agreement to single-device randomness. Manuscript. [Anonymous Institution], Oslo, Norway. Target: ACM WiSec 2026, 2026.

[5] Melih Aslan, Ali Doganaksoy, and Zubeyde Saygi. Observations on NIST SP 800-90B entropy estimators. *Cryptography and Communications*, 2025.

[6] Elaine Barker and John Kelsey. Recommendation for random bit generator (RBG) constructions. Technical Report SP 800-90C, NIST, 2025.

[7] Katharina Boudgoust and Oleksandra Lapiha. Leftover hash lemma(s) over cyclotomic rings, 2025. IACR ePrint 2025/1080.

[8] J. Lawrence Carter and Mark N. Wegman. Universal classes of hash functions. *J. Comput. Syst. Sci.*, 18(2):143–154, 1979.

[9] Eshan Chattopadhyay and Jesse Goodman. Leakage-resilient extractors against number-on-forehead protocols. In *STOC 2025*, 2025. arXiv:2506.12595.

[10] Sandro Coretti, Pooya Farshim, Patrick Harasser, and Luke Southern. Multi-source randomness extraction and generation in the random-oracle model. In *ITC 2025*, 2025.

[11] Vo Thi Anh Dang et al. Entropy extractor based high-throughput post-processings for TRNGs. *IACR TCHES*, 2025(4):145–171, 2025.

[12] Itai Dinur. Tight indistinguishability bounds for the XOR of independent random permutations by Fourier analysis. In *EUROCRYPT 2024*, 2024. IACR ePrint 2024/338.

[13] Yevgeniy Dodis, Rafail Ostrovsky, Leonid Reyzin, and Adam Smith. Fuzzy extractors: How to generate strong keys from biometrics and other noisy data. *SIAM J. Computing*, 38(1):97–139, 2008.

[14] Dean Doron and João Ribeiro. Nearly-linear time seeded extractors with short seeds, 2024. arXiv:2411.07473.

[15] Morris J. Dworkin. SHA-3 standard: Permutation-based hash and extendable-output functions. Technical Report FIPS 202, NIST, 2015.

[16] European Parliament and Council. Regulation (EU) 2022/2554 on digital operational resilience for the financial sector (DORA), 2022. Official Journal of the European Union, L 333, 27 Dec 2022. Effective 17 Jan 2025 (Norway: 1 July 2025).

[17] Cameron Foreman and Llúís Masanes. Seedless extractors for device-independent quantum cryptography. *Quantum*, 9:1654, 2025. arXiv:2403.04713.

[18] Cameron Foreman, Richie Yeung, Alec Edgington, and Florian J. Curchod. Cryptomite: A versatile and user-friendly library of randomness extractors, 2024. arXiv:2402.09481.

[19] Louis Golowich. New explicit constant-degree lossless expanders. In *SODA 2024*, 2024. arXiv:2306.07551.

[20] Jesse Goodman, Xin Li, and David Zuckerman. Improved condensers for Chor–Goldreich sources. In *FOCS 2024*, 2024. arXiv:2410.08142.

[21] Milos Grujic and Ingrid Verbauwhede. Optimizing linear correctors: A tight output min-entropy bound and selection technique. *IEEE Trans. Information Forensics and Security*, 2024. arXiv:2304.05306.

[22] Venkatesan Guruswami, Christopher Umans, and Salil Vadhan. Unbalanced expanders and randomness extractors from Parvaresh–Vardy codes. In *J. ACM*, volume 56, 2009.

[23] Russell Impagliazzo, Leonid A. Levin, and Michael Luby. Pseudorandom generation from one-way functions. In *Proc. 21st ACM Symp. Theory of Computing (STOC)*, pages 12–24, 1989.

[24] Hanjun Jiang, Yongge Wang, Song Luo, Jintai Liu, Yue Gang, Yanbin Pan, and Yuyu Wang. Re-randomize and extract: Novel commitment construction based on group actions. In *Eurocrypt 2025*, 2025.

[25] Minzhao Liu, Ruslan Shaydulin, Pradeep Niroula, Matthew DeCross, Shih-Han Hung, Scott Aaronson, et al. Certified randomness using a trapped-ion quantum processor. *Nature*, 640:343–348, 2025. arXiv:2503.20498.

[26] Dafu Lou et al. Raw QPP-RNG randomness via system jitter across platforms: A NIST SP 800-90B evaluation. *Scientific Reports*, 2025.

[27] Xiongfeng Ma, Xiao Yuan, Zhu Cao, Bing Qi, and Zhen Zhang. Quantum random number generation. *npj Quantum Information*, 2:16021, 2016.

[28] Kerry McKay, Werner Schindler, and Meltem Sönmez Turan. Bridging the gap between standards on random number generation. Technical Report NIST IR 8446, NIST/BSI, 2026.

[29] Ralph C. Merkle. A certified digital signature. *Advances in Cryptology (CRYPTO)*, pages 218–238, 1989.

[30] National Institute of Standards and Technology. Module-lattice-based key-encapsulation mechanism standard. Technical Report FIPS 203, NIST, 2024.

[31] Michael Sandfuchs, Marco Ferradini, and Renato Renner. Randomness from causally independent processes, 2025. arXiv:2510.05203.

[32] Werner Schindler and Matthias Peter. A proposal for functionality classes for random number generators, version 3.0. Technical report, BSI, 2024.

[33] Luca Trevisan. Extractors and pseudorandom generators. In *J. ACM*, volume 48, pages 860–879, 2001.

[34] Meltem Sönmez Turan, Elaine Barker, John Kelsey, Kerry A. McKay, Mary L. Baish, and Mike Boyle. Recommendation for the entropy sources used for random bit generation. Technical Report SP 800-90B, NIST, 2018.